

Tutorial 1

**Distributed Computing
(COMP3008, COMP4002, COMP5002)**

**Discipline of Computing School of Electrical
Engineering, Computing and Mathematical
Sciences (EECMS)**

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Provide Code: 00301J**

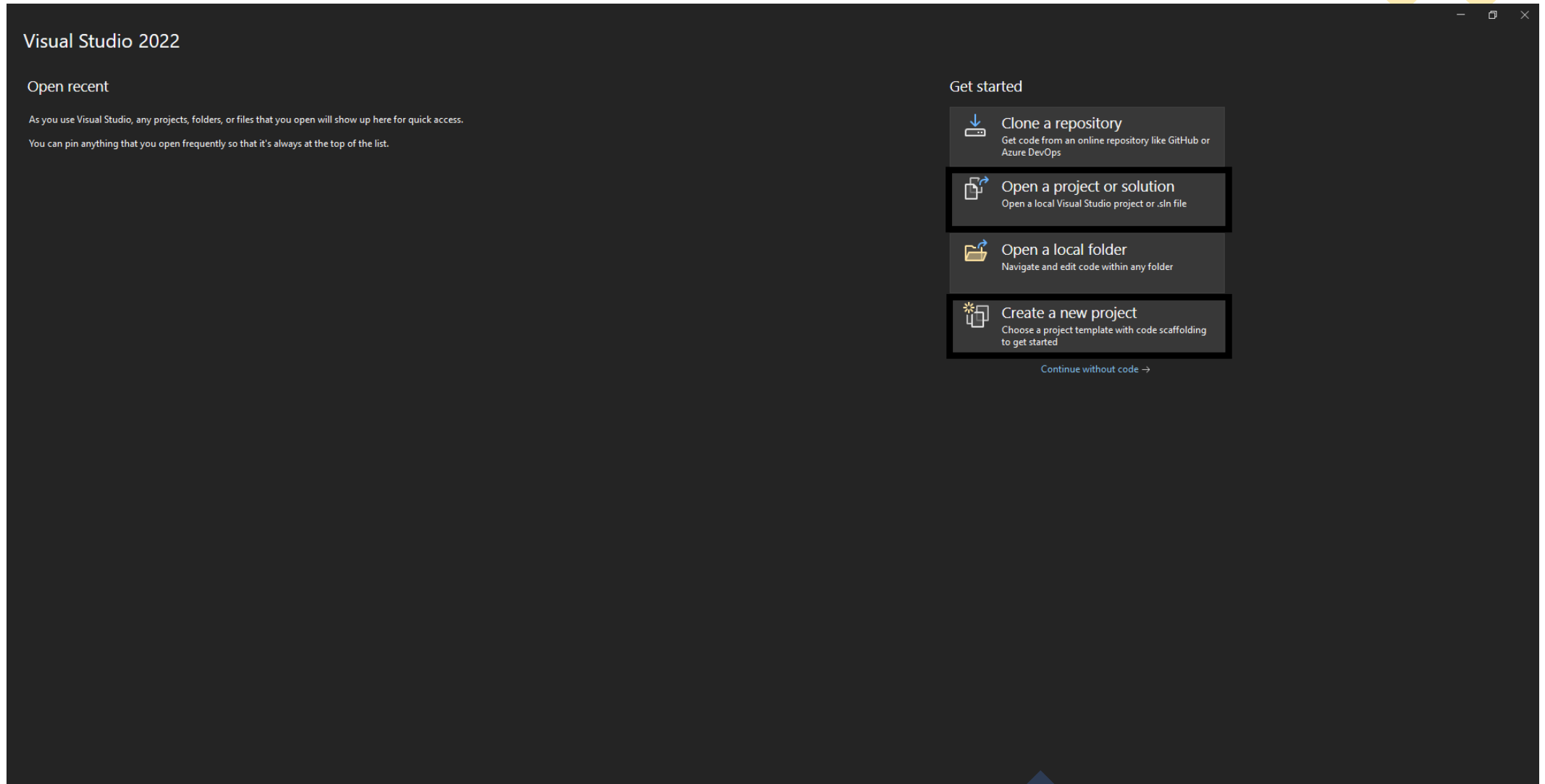


Objectives

- **Console application – C#**
- **WPF Application**
- **Study Lecture 1**
- **Check the source code attached to Lecture 1 – WinAPPFirst.zip**
- **You can start your projects from scratch or modify the project/source codes given in Lecture 1**
- **Check the due date on the last slide.**



The next slides reiterate some of the Lecture 1 slides. If you understand how to build console and WPF projects, you can directly go to slide 33, which specifies the task/modification required.



Create a new project

Recent project templates

 Console App (.NET Framework)

C#

Search for templates (Alt+S)



[Clear all](#)

C#

Windows

Console



Console App

A project for creating a command-line application that can run on .NET on Windows, Linux and macOS

C#

Linux

macOS

Windows

Console



Console App (.NET Framework)

A project for creating a command-line application

C#

Windows

Console

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Configure your new project

Console App (.NET Framework)

C#

Windows

Console

Project name

ConsoleApp1

Location

C:\Users\290574c\source\repos

Solution name ⓘ

ConsoleApp1

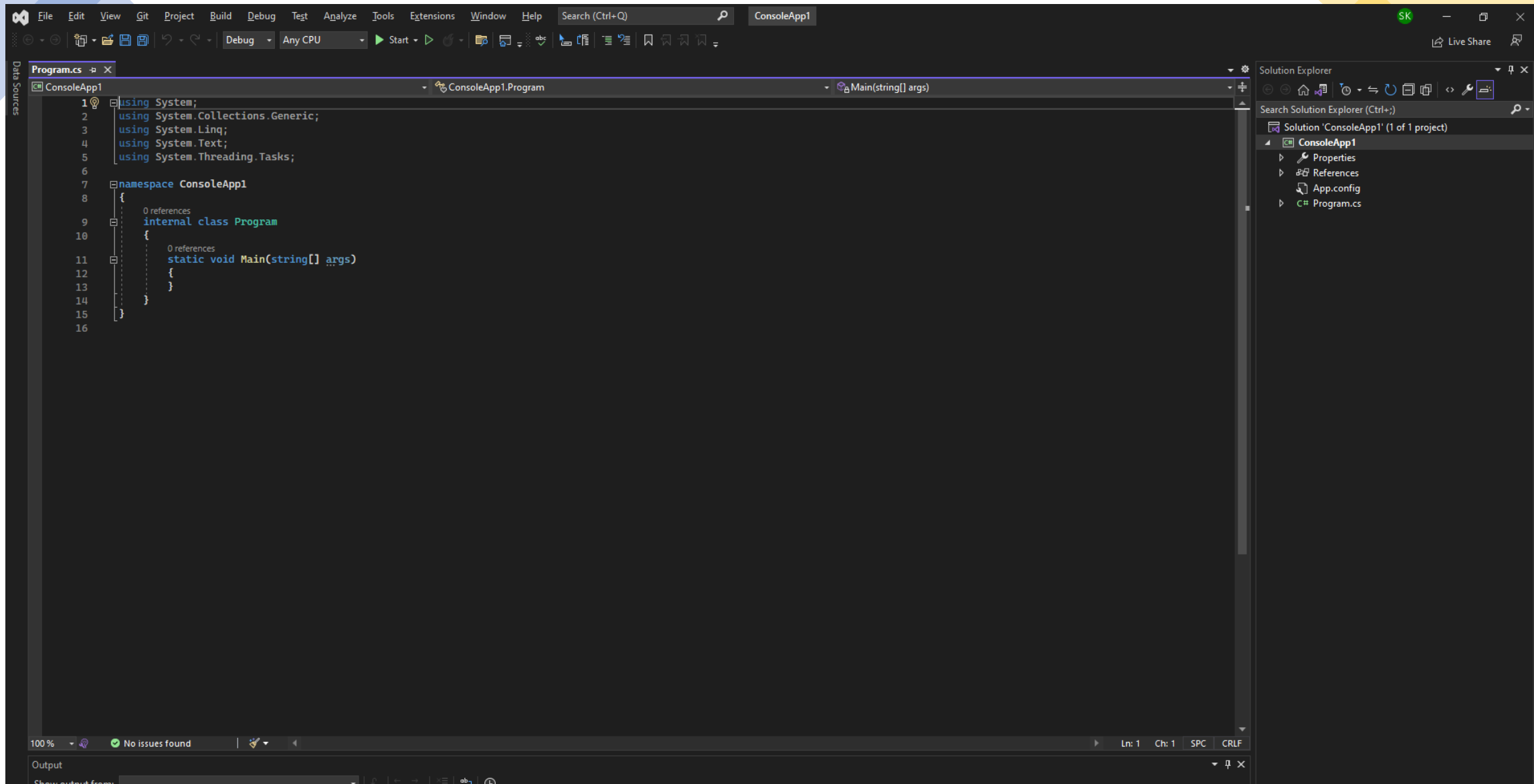
☐

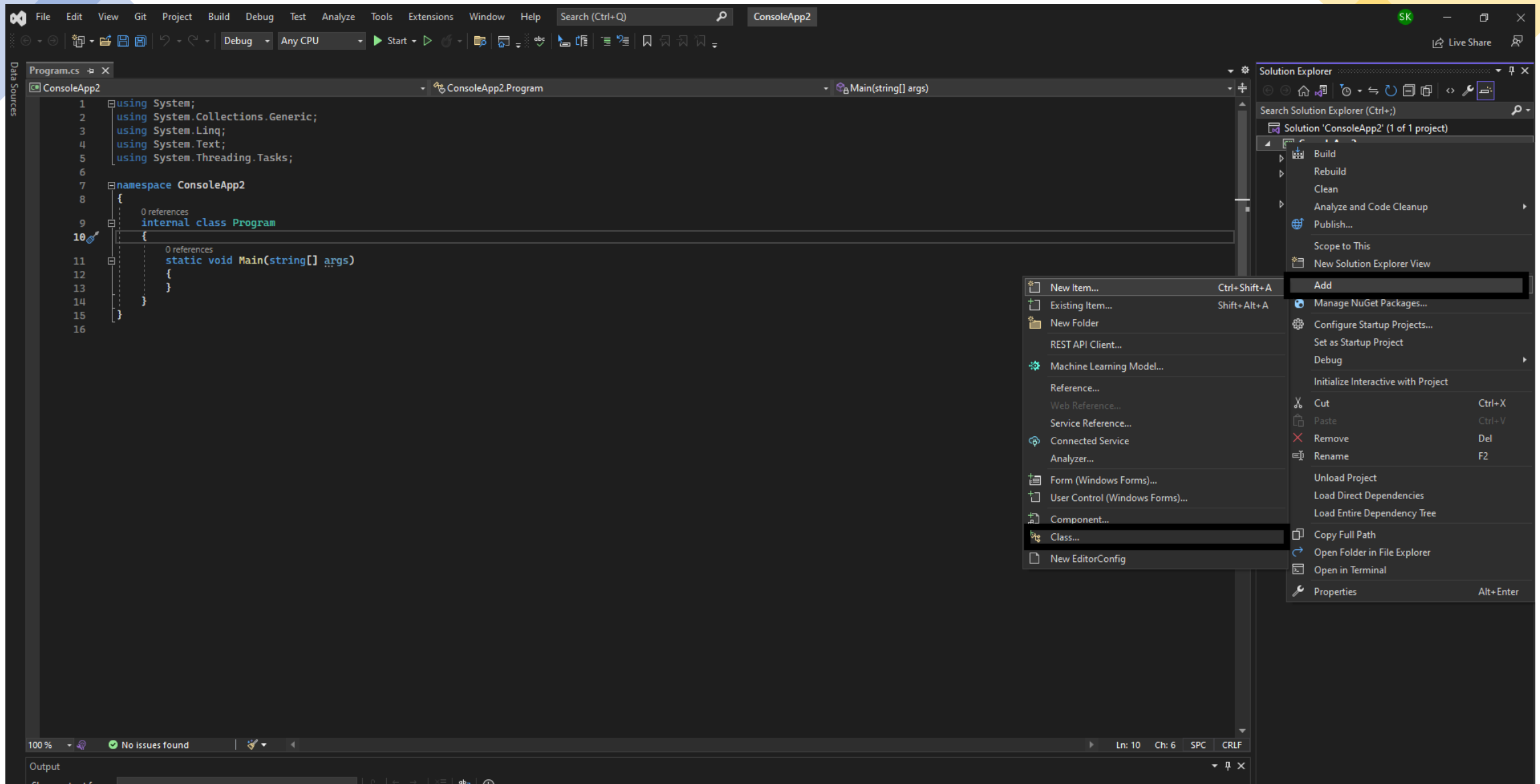
Place solution and project in the same directory

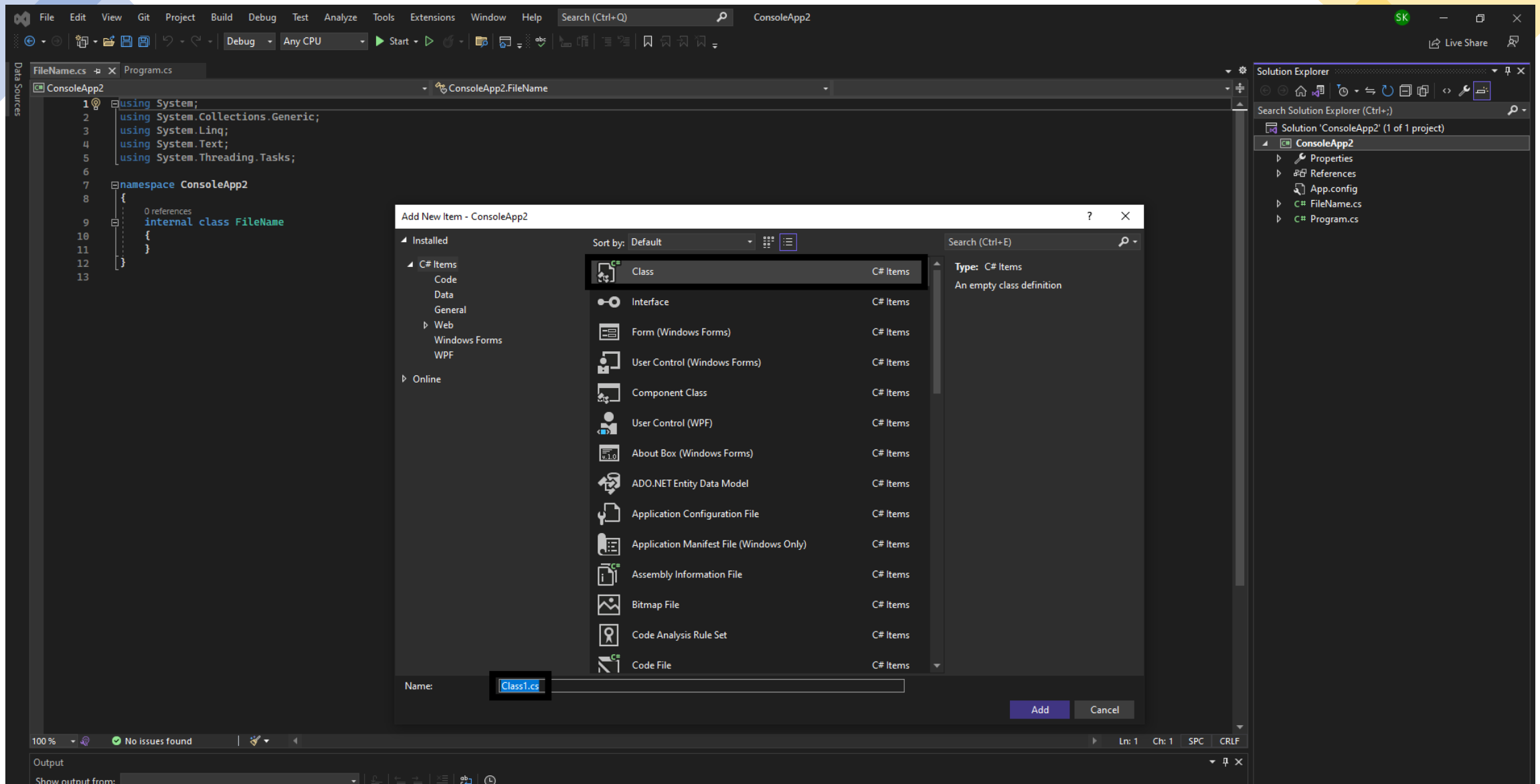
Framework

.NET Framework 4.7.2

Project will be created in "C:\Users\290574c\source\repos\ConsoleApp1\ConsoleApp1\."







Visual Studio interface showing a C# console application named ConsoleApp2. The main code file, Program.cs, contains the following code:

```
1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Text;
5 using System.Threading.Tasks;
6
7 namespace ConsoleApp2
8 {
9     0 references
10     internal class Program
11     {
12         0 references
13         static void Main(string[] args)
14         {
15             Console.WriteLine("Hello word");
16             Console.ReadLine();
17         }
18 }
```

The Solution Explorer on the right shows the project structure:

- Solution 'ConsoleApp2' (1 of 1 project)
 - ConsoleApp2
 - Properties
 - References
 - App.config
 - Class1.cs
 - FileName.cs
 - Program.cs

The Error List at the bottom shows two errors:

Code	Description	Project	File	Line	Suppression State
CS1519	Invalid token ']' in class, record, struct, or interface member declaration	ConsoleApp2	Class1.cs	12	Active
CS1002	; expected	ConsoleApp2	Program.cs	13	Active

```
1  using System;
2  using System.Collections.Generic;
3  using System.Linq;
4  using System.Text;
5  using System.Threading.Tasks;
6
7  namespace ConsoleApp1
8  {
9      1 reference
10     public class Person
11     {
12         private string name;
13         7 references
14         public string Name {
15             get { return name; }
16             set { name = value; }
17         }
18     }
19 }
```

7 namespace ConsoleApp1

8 {

19 references

9 public class Student: Person

10 {

11 private int id;

12 private string university;

13

7 references

14 public int Id

15 {

16 get { return id; }

17 set { id = value; }

18 }

6 references

19 public string University

20 {

21 get { return university; }

22 set { university = value; }

23 }

24

1 reference

25 public override string ToString()

26 {

27 string info = "The student's name is " + Name + "\n";

28 info=info + "The student's id is " + id + "\n";

29 info = info+ "The student's university is " + university;

30 return info;

31 }

32

33 }

34 }

C# ConsoleApp1

ConsoleApp1.StudentList

```
7 namespace ConsoleApp1
8 {
9     3 references
10    public class StudentList
11    {
12        3 references
13        public static List<Student> Students()
14        {
15            List<Student> slist = new List<Student>();
16
17            Student student1 = new Student();
18            student1.Name = "Robert";
19            student1.University = "Curtin";
20            student1.Id = 101;
21
22            Student student2 = new Student();
23            student2.Name = "Mia";
24            student2.University = "EWU";
25            student2.Id = 102;
26
27            Student student3 = new Student();
28            student3.Name = "Adam";
29            student3.University = "Murdoch";
30            student3.Id = 103;
31
32            slist.Add(student1);
33            slist.Add(student2);
34            slist.Add(student3);
35
36            return slist;
37        }
38    }
39 }
40
41 }
```

```
1  using System;
2  using System.Collections.Generic;
3  using System.Linq;
4  using System.Text;
5  using System.Threading.Tasks;
6
7  namespace ConsoleApp1
8  {
9      0 references
9      internal class Program
10     {
11         0 references
11         static void Main(string[] args)
12         {
13             List<Student> studentlist = StudentList.Students();
14
15             foreach (Student student in studentlist)
16             {
17                 Console.WriteLine("*****");
18                 Console.WriteLine(student);
19             }
20         }
21         Console.WriteLine("*****");
22         Console.ReadLine();
23     }
24 }
25
```

Visual Studio IDE showing the development and execution of a C# console application named ConsoleApp1.

Code Editor (Program.cs):

```
1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Text;
5 using System.Threading.Tasks;
6
7 namespace ConsoleApp1
8 {
9     0 references
10     internal class Program
11     {
12         0 references
13         static void Main(string[] args)
14         {
15             List<Student> studentlist = StudentList.Students();
16
17             foreach (Student student in studentlist)
18             {
19                 Console.WriteLine("*****");
20                 Console.WriteLine(student);
21             }
22             Console.WriteLine("*****");
23             Console.ReadLine();
24         }
25     }
```

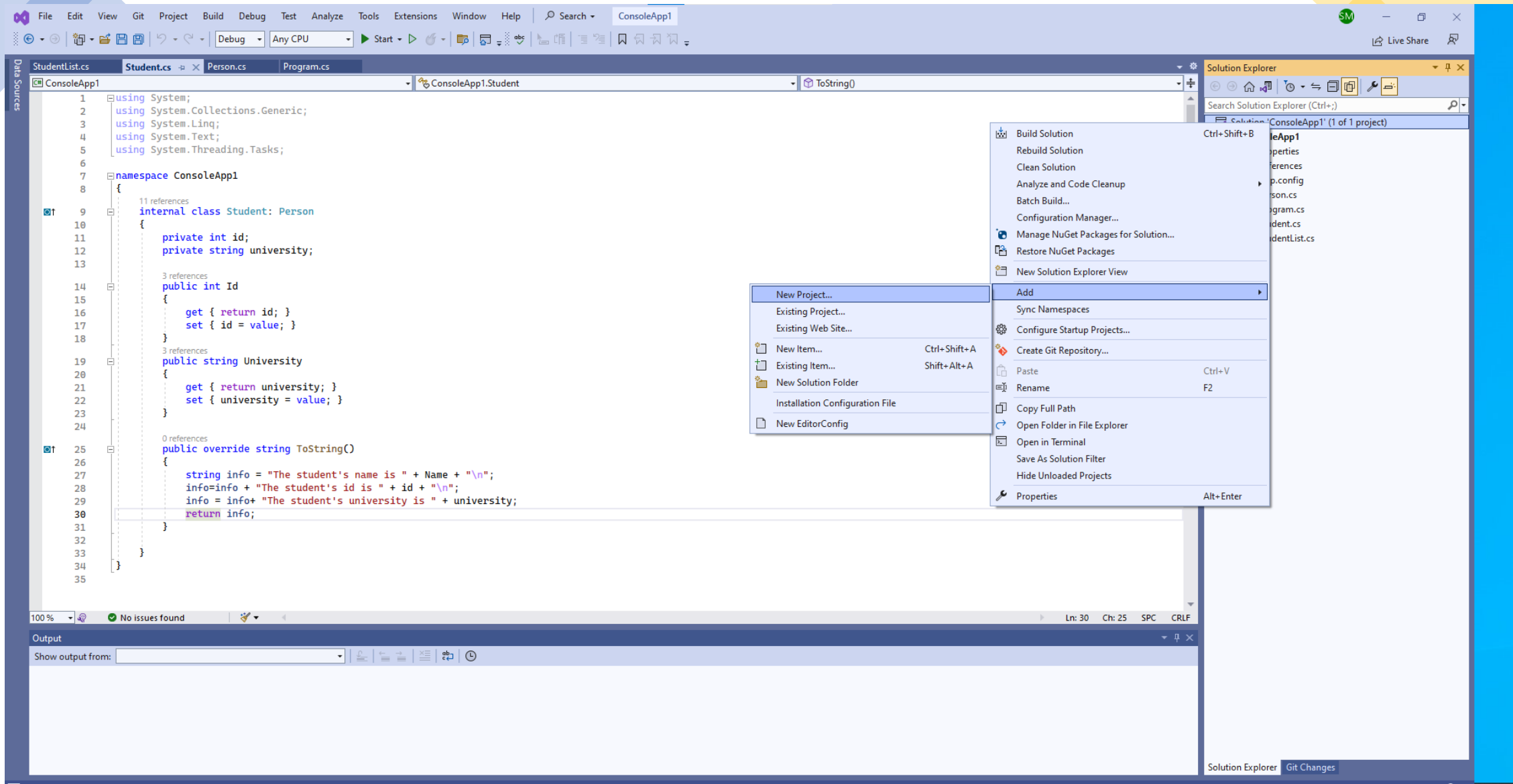
Output Window:

```
C:\Users\290574c\Downloads\ConsoleApp1 (1)\ConsoleApp1\ConsoleApp1\bin\Debug\ConsoleApp1.exe
*****
The student's name is Robert
The student's id is 101
The student's university is Curtin
*****
The student's name is Mia
The student's id is 102
The student's university is EWU
*****
The student's name is Adam
The student's id is 103
The student's university is Murdoch
*****
```

Bottom Panel:

- Autos:** Search (Ctrl+E), Search Depth: [dropdown]
- Output:** Show output from: Debug. Log entries for ConsoleApp1.exe (CLR v4.0.30319; DefaultDomain): Loaded 'C:\Windows\Microsoft.Net' and 'C:\Users\290574c\Downloa'.
- Input Switch Notification:** Input language switching. Typing Left Alt + Shift changes your input language. You can turn this feature off or change your hot key sequence by selecting Customise.
- Buttons:** Customise, Dismiss

Footer: Autos | Locals | Watch 1 | Call Stack | Breakpoints | Exception Settings | Command Window | Immediate Window | Output



Add a new project

Recent project templates

A list of your recently accessed templates will be displayed here.

✕ ▼ [Clear all](#)

All languages ▼

All platforms ▼

All project types ▼

 **WPF Application**
A project for creating a .NET **WPF** Application
C# Windows Desktop

 **WPF Application**
A project for creating a .NET **WPF** Application
Visual Basic Windows Desktop

 **WPF Class Library**
A project for creating a class library that targets a .NET **WPF** Application
C# Windows Desktop Library

 **WPF Class Library**
A project for creating a class library that targets a .NET **WPF** Application
Visual Basic Windows Desktop Library

 **WPF Custom Control Library**
A project for creating a custom control library for .NET **WPF** Applications
C# Windows Desktop Library

 **WPF Custom Control Library**
A project for creating a custom control library for .NET **WPF** Applications
Visual Basic Windows Desktop Library

[Next](#)

Configure your new project

WPF Application

C#

Windows

Desktop

Project name

WpfApp1

Location

C:\Users\283602K\Downloads\DC Materials\Lecture-1\DC_Lecture1_Demo\ConsoleApp1

...

Project will be created in "C:\Users\283602K\Downloads\DC Materials\Lecture-1\DC_Lecture1_Demo\ConsoleApp1\WpfApp1\"

Back

Next

Additional information

WPF Application

C#

Windows

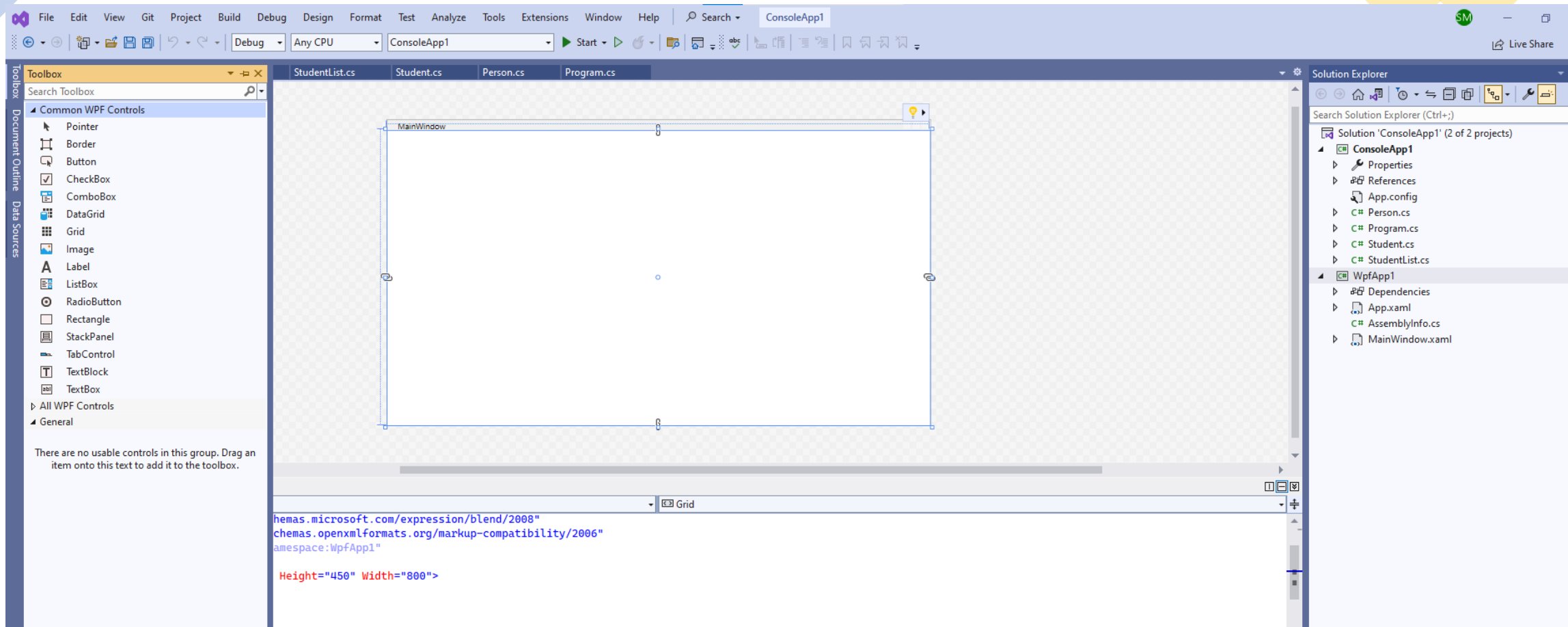
Desktop

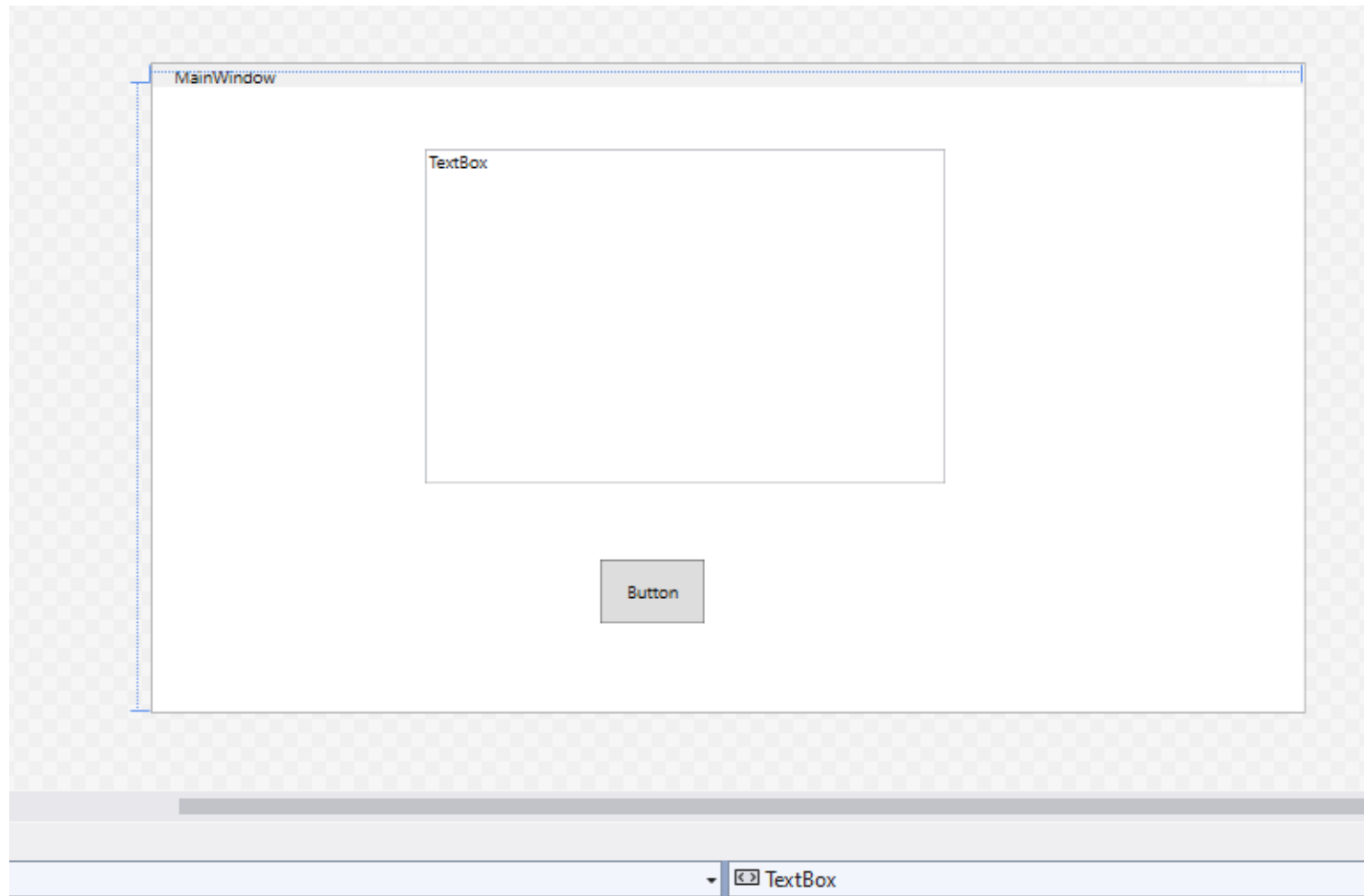
Framework 

.NET 6.0 (Long Term Support)

Back

Create





openxmlformats.org/markup-compatibility/2006"

WpfApp1"

="450" Width="800">

ent="Left" Margin="189,43,0,0" TextWrapping="Wrap" Text="TextBox" VerticalAlignment="Top" Width="362"

MainWindow

328

311

GET ALL

pace:WpfApp1"

ght="450" Width="800">

gnment="Left" Margin="189,43,0,0" TextWrapping="Wrap" VerticalAlignment="Top" Width="362" Height="232"/>

LL" HorizontalAlignment="Left" Margin="311,328,0,0" VerticalAlignment="Top" Height="45" Width="73" RenderTransformOrigin="1.14,0.449" Click="Button_Click"/>

Ln: 11 Ch: 16 SPC CRLF

Search Solution Explorer (Ctrl+)

Solution 'ConsoleApp1' (2 of 2 projects)

- ConsoleApp1
 - Properties
 - References
 - App.config
 - Person.cs
 - Program.cs
 - Student.cs
 - StudentList.cs
- WpfApp1
 - Dependencies
 - App.xaml
 - AssemblyInfo.cs
 - MainWindow.xaml

Solution Explorer Git Changes

Properties

Name <No Name>

Type Button

Arrange by: Category

- Brush
- Layout
- Text
- Appearance

Common

Command

CommandPara...

Content GET ALL

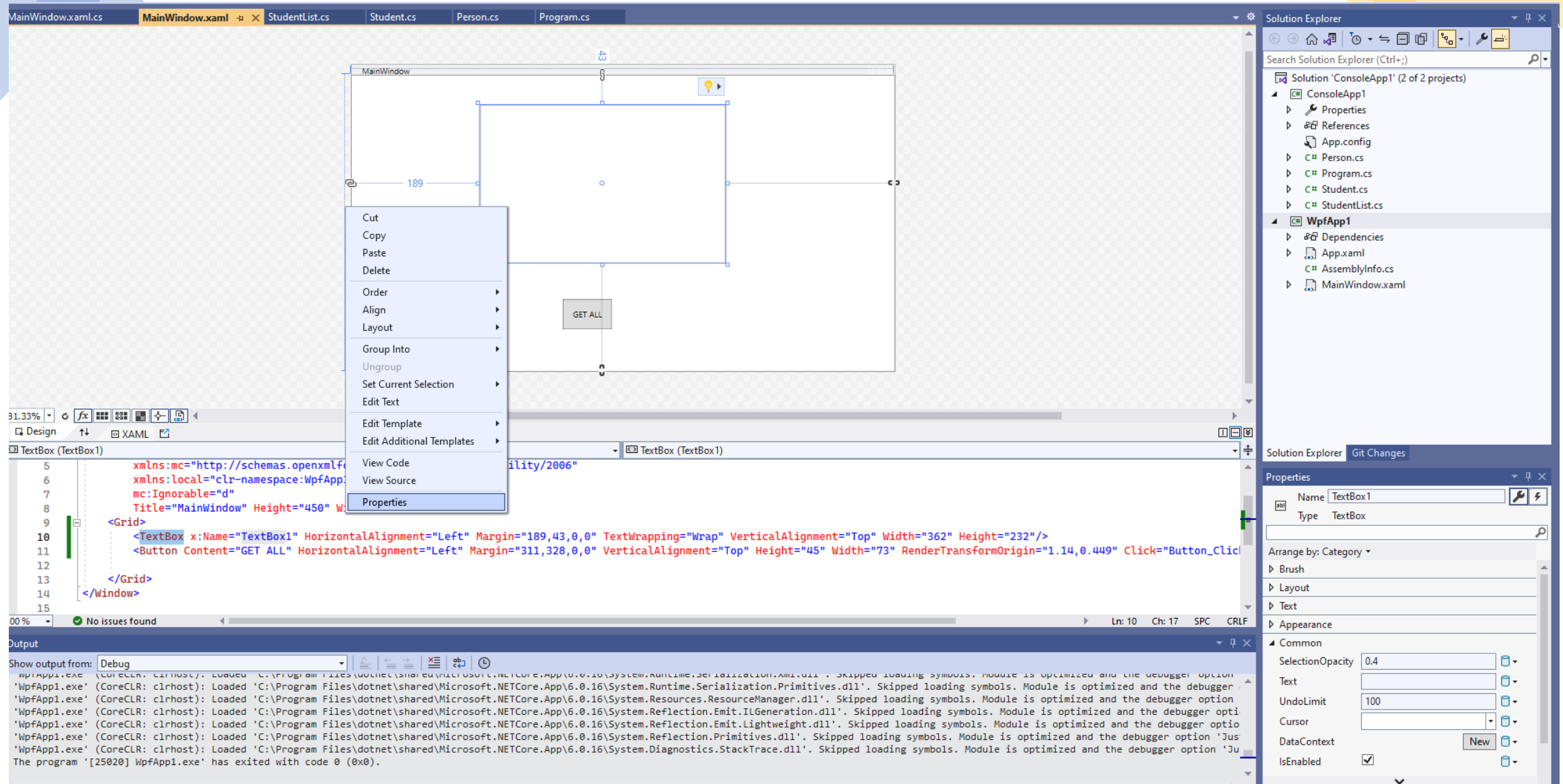
IsCancel

IsDefault

Cursor

DataContext

New



```
using System.Windows.Shapes;
```

```
namespace WpfApp1
```

```
{
```

```
    /// <summary>
```

```
    /// Interaction logic for MainWindow.xaml
```

```
    /// </summary>
```

```
    2 references
```

```
    public partial class MainWindow : Window
```

```
    {
```

```
        0 references
```

```
        public MainWindow()
```

```
        {
```

```
            InitializeComponent();
```

```
        }
```

```
        1 reference
```

```
        private void Button_Click(object sender, RoutedEventArgs e)
```

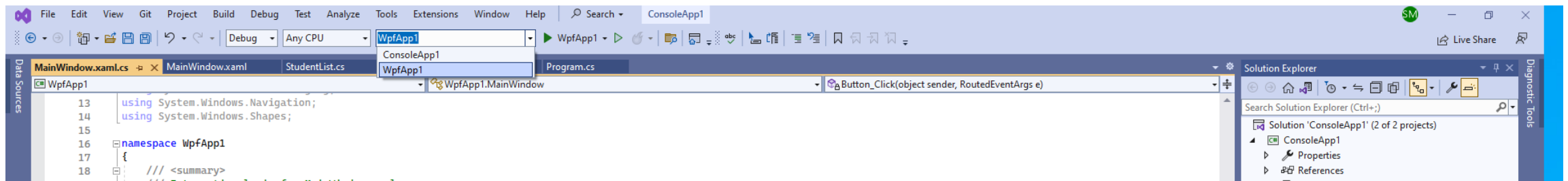
```
        {
```

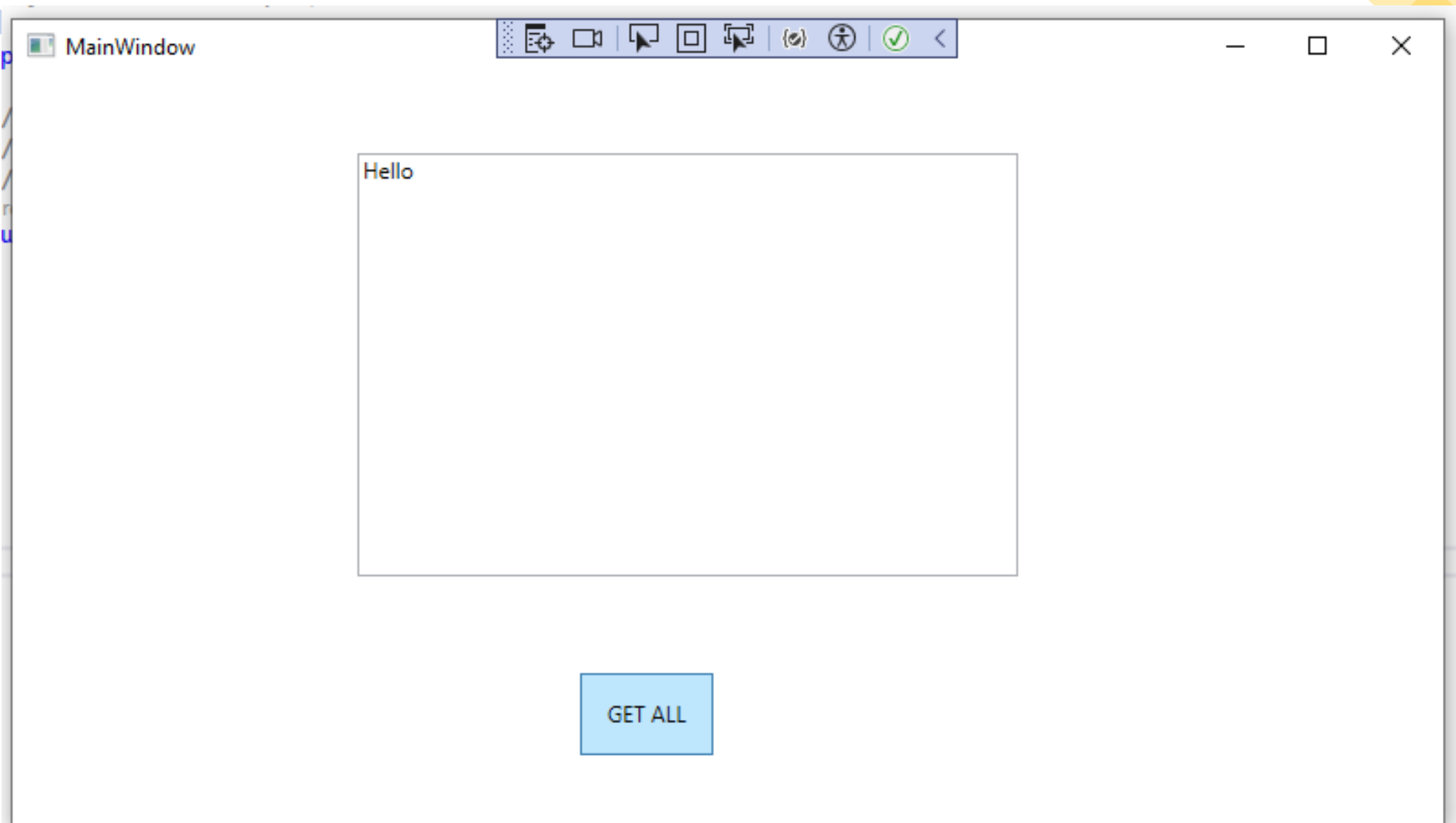
```
            TextBox1.Text = "Hello";
```

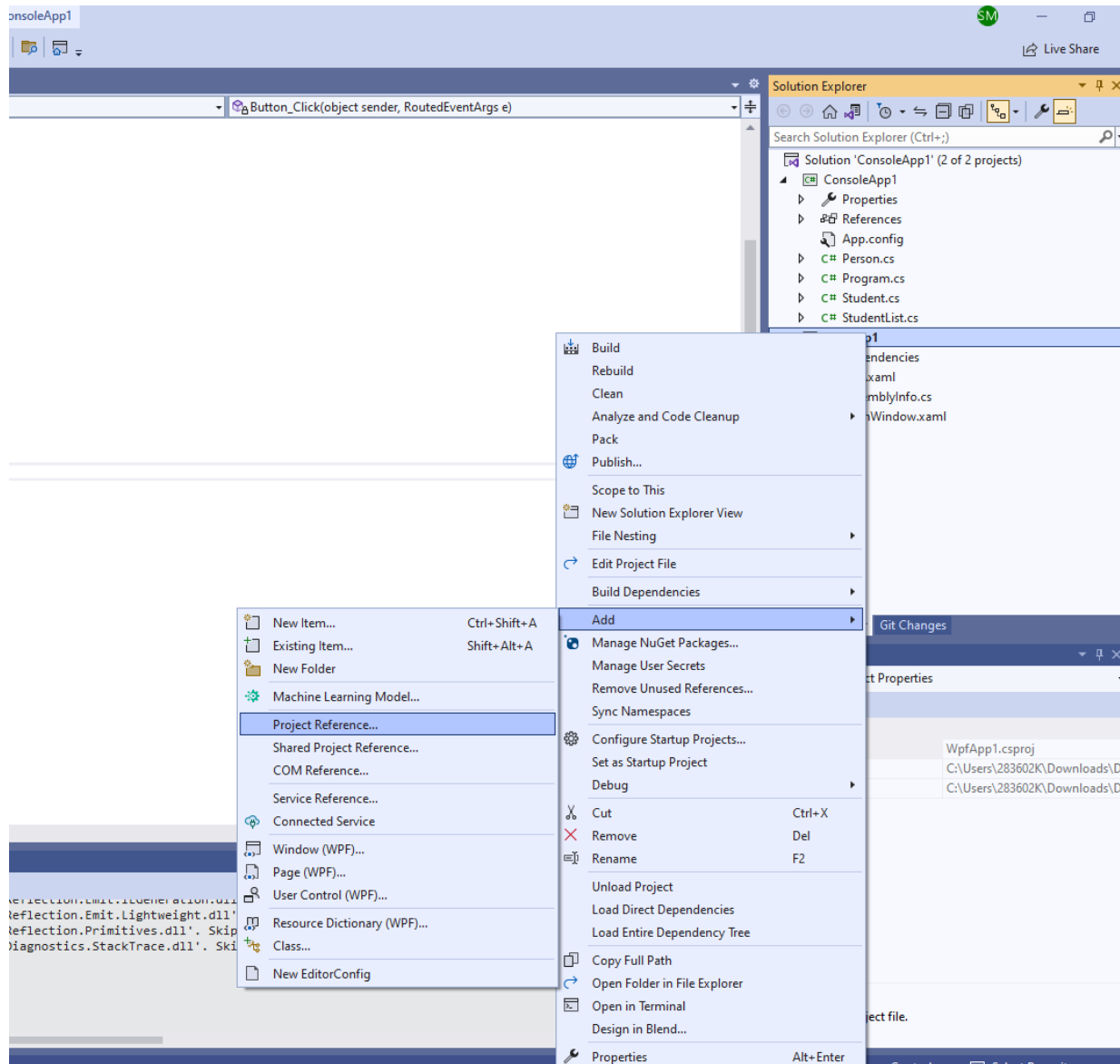
```
        }
```

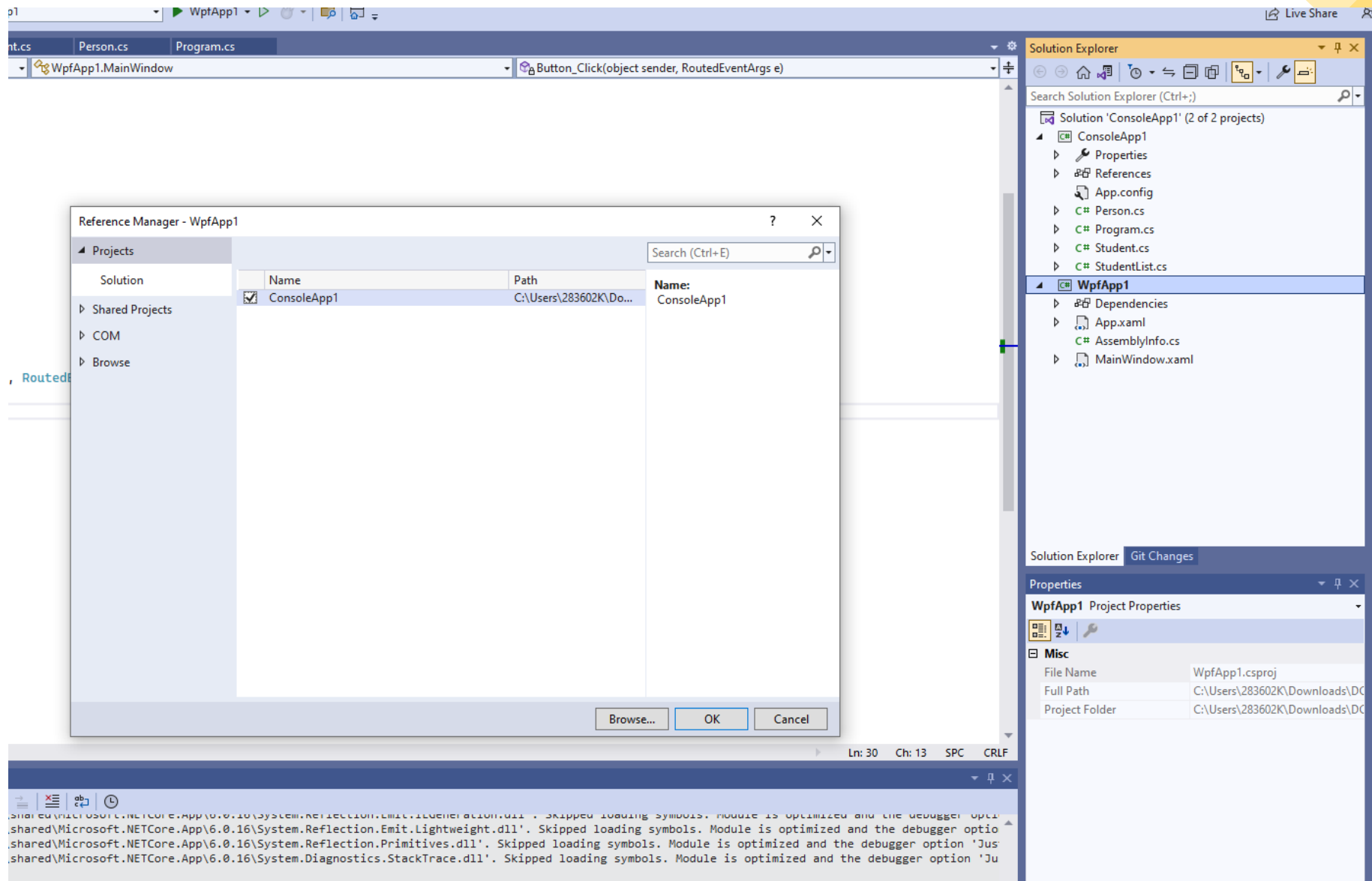
```
    }
```

```
}
```









```
using ConsoleApp1;

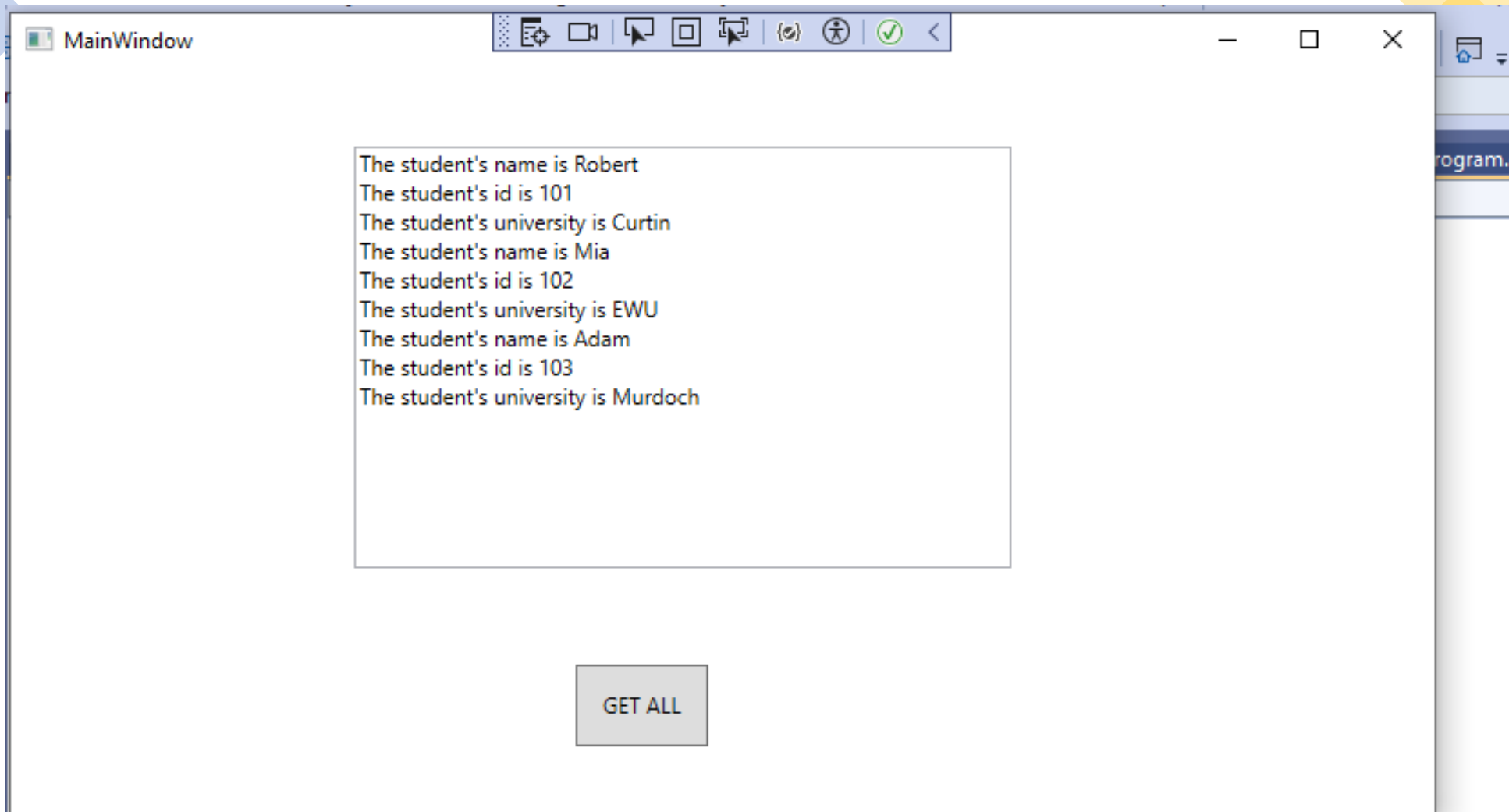
namespace WpfApp1
{
    /// <summary>
    /// Interaction logic for MainWindow.xaml
    /// </summary>
    2 references
    public partial class MainWindow : Window
    {
        0 references
        public MainWindow()
        {
            InitializeComponent();
        }

        1 reference
        private void Button_Click(object sender, RoutedEventArgs e)
        {
            string value = "";

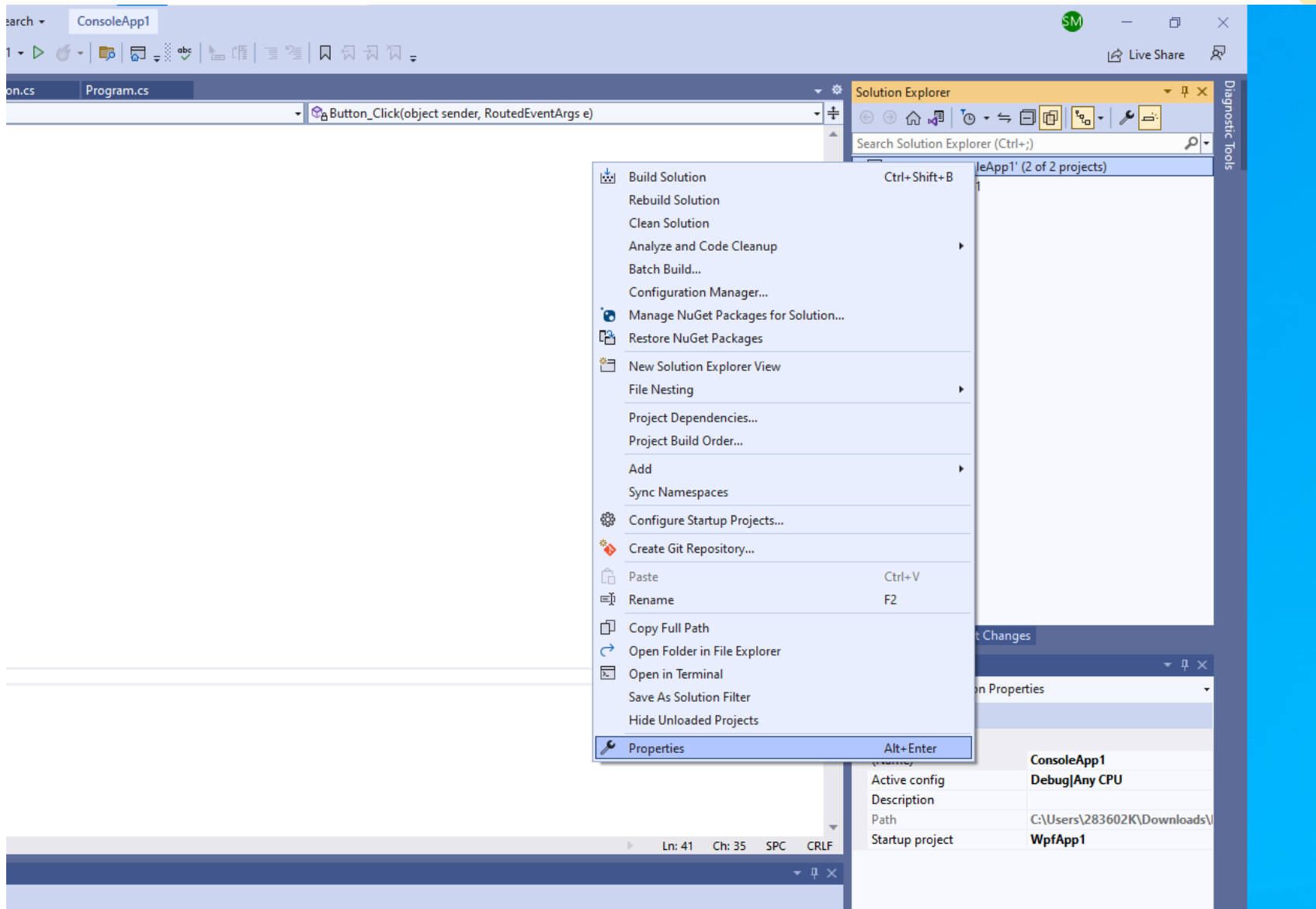
            List<Student> studentlist = StudentList.Students();

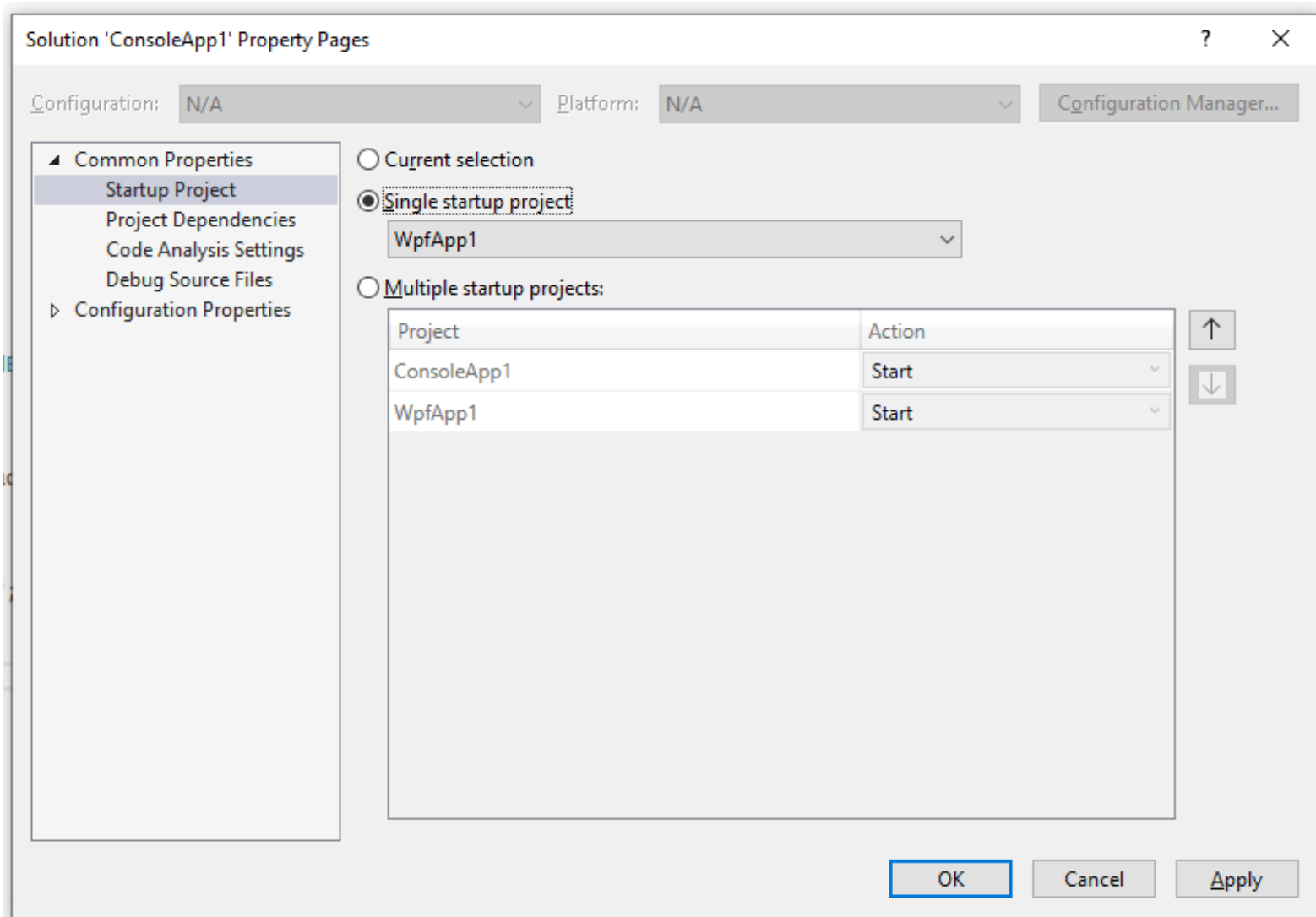
            foreach (Student student in studentlist)
            {
                value = value + student.ToString()+"\n";
            }

            TextBox1.Text = value;
        }
    }
}
```



```
30 private void Button_Click(object sender, RoutedEventArgs e)
31 {
32     string value = "";
```





Visual Studio IDE showing a WPF application (WpfApp1) running in debug mode. The application is displaying a list of student records in a console window and a dialog box.

Console Output:

```
*****
The student's name is Robert
The student's id is 101
The student's university is Curtin
*****
The student's name is Mia
The student's id is 102
The student's university is EWU
*****
The student's name is Adam
The student's id is 103
The student's university is Murdoch
*****
```

Dialog Box Content:

```
The student's name is Robert
The student's id is 101
The student's university is Curtin
The student's name is Mia
The student's id is 102
The student's university is EWU
The student's name is Adam
The student's id is 103
The student's university is Murdoch
```

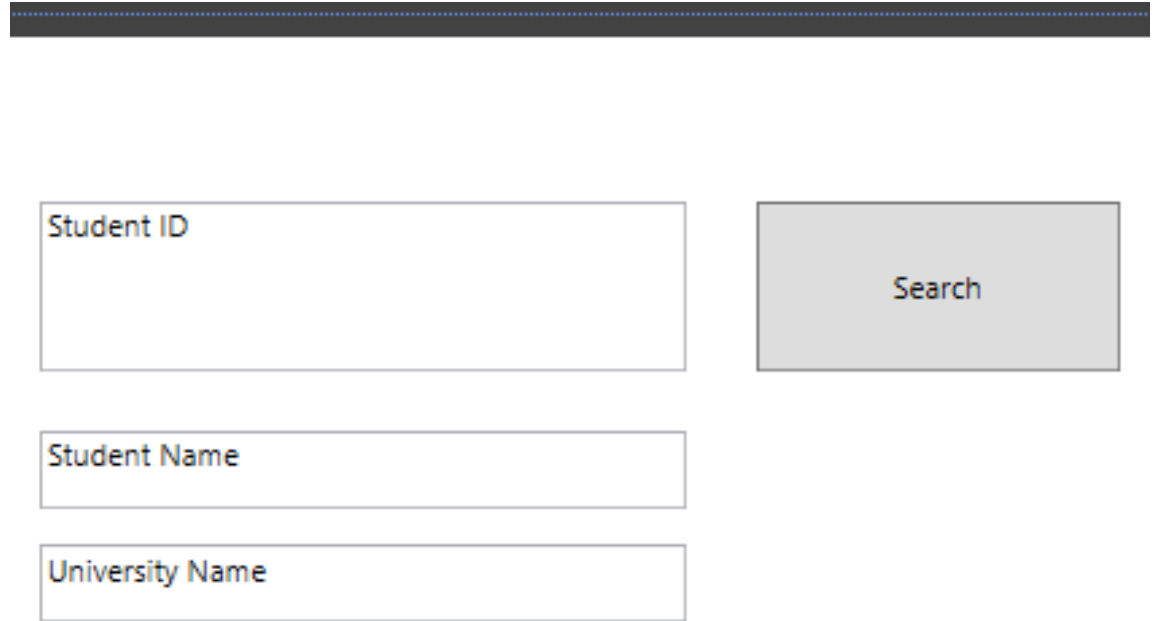
GET ALL

Call Stack:

Name	Value	Type
Search (Ctrl+E)		

Expected Output

- If you've made it here, you now know how to build WPF application.
- Add a couple of more student information in the student.
- Try finding the student's name by passing the student id as input.



The image shows a WPF application window with a dark title bar. Inside the window, there are three text input fields stacked vertically on the left side. The first field is labeled "Student ID", the second is labeled "Student Name", and the third is labeled "University Name". To the right of these fields is a rectangular button labeled "Search".